

The Crossing Number of the Hypercube

OPG-313	L1 - Tractable	Ouvert	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

The crossing number $\text{cr}(G)$ of G is the minimum number of crossings in all drawings of G in the plane.

The d -dimensional (hyper)cube Q_d is the graph whose vertices are all binary sequences of length d , and two of the sequences are adjacent in Q_d if they differ in precisely one coordinate.

Conjecture $\displaystyle \lim \frac{\text{cr}(Q_d)}{4^d} = \frac{5}{32}$

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